

# Complement frames and sharing across common-point groups

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## Abstract

Orthogonal complements of forward source spans supply the reversed-stage frames, allowing equal intermediate sums from different common-point groups to be shared. The new conditional saving is  $\kappa = 13 \cdot 2^{-66}$ , a factor 13/8 above the preceding  $2^{-63}$  witness. This is a small refinement and a reusable transfer lemma, not an exponent in the 50s or an optimality claim for general finite networks.

## 1 Scope

This note extends the preceding shared-computation and rectangle constructions. It retains their reversible embedding, central network, and first/third-stage auxiliary matching, and the unchanged interfaces of OpenAI’s *Integer multiplication below  $n \log n$* , pinned at

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The fixed finite-alphabet, fixed finite number of one-dimensional tapes model, complex network, numerical maps, and rounding are unchanged. The accompanying independent upstream patch includes the preceding constructions in full.

### 1.1 Sharing sums across different common points

Retain  $h = 46$ ,  $F = \mathbb{Q}^h$  with form  $I - J/9$ ,  $v = 15180$ ,  $N = v^3$ , and  $m = 97336$ . The following extension of the preceding computation allows equal sums from different common-point groups to be identified.

**Proposition 1** (Complement frames for the reversed computation). *Consider a cancellation-free binary-addition graph, embedded reversibly by reusing a pivot along one outgoing edge at each addition. Suppose the source indicator span  $U_z$  at every node  $z$  is nondegenerate, and every source contributing to a designated output is orthogonal to that output’s target indicator. Then the side computation admits increasing nondegenerate frames in both stage orientations, with no extra decreasing edges.*

*Proof.* In the forward middle computation, use  $D_{U_z} = (B \otimes F) \perp (P \otimes U_z)$ , with the future nondegenerate line factor suppressed. Along every directed edge  $z \rightarrow w$ ,  $U_z \subset U_w$ . Physical roles follow paths, so these labels increase. At an output,  $U_z \subset t_T^\perp$ , as required by the target injection.

In the reversed middle computation, use  $D_{U_z^\perp}$  instead. The ambient form is nondegenerate, so the orthogonal complement of every nondegenerate  $U_z$  is nondegenerate. Reversing an edge gives

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$U_w^\perp \subset U_z^\perp$ . At an original output, the physical  $X$  line satisfies  $\langle t_T \rangle \subset U_z^\perp$ ; at an original input, the final complement is exactly the physical  $Y$  source complement  $t_S^\perp$ . These are the required reversed-stage endpoints. All roles incident to a node gate receive the same frame. Newly active roles enter from  $D_0 = B \otimes F$  and retired roles grow to  $D_1 = A \otimes F$ . Early mixers use  $D_0$  and late cleanup mixers use  $D_1$ , as before. Tensoring with the future line preserves these conclusions. Thus all side edges are increasing in both orientations.  $\square$

This argument does not require the reachable-target spans to be positive definite. It also does not permit arbitrary equal-value circuit rewrites: nondegeneracy of the source spans and the stated endpoint orthogonality remain necessary hypotheses of this transfer.

Start with the preceding  $h$  copies of the  $n = 45$  pair-exclusion circuit. Identify their input nodes when they name the same physical triple. Identify addition nodes whenever their formal source-support sets agree, retaining the first encountered decomposition. Process common points in increasing order and the local node identifiers in increasing order, then discard unused nodes. Keep all  $q = h \binom{h-1}{2} = 45540$  designated partial outputs. Their injection into the physical target bank adds the three contributions for each target triple. A neighboring pair has a unique common point, so the resulting matrix remains precisely the intersection-one relation.

There is a compact exact method for finding every cross-group equality. If a sum belongs to groups  $i \neq j$ , every source triple contains both  $i$  and  $j$ . Its support is therefore a pair-star  $\{\{i, j, k\} : k \in K\}$ , or a single input triple. The fixed pair and the union of its vertices uniquely determine this support. All other addition nodes belong to a single group and were already interned in the local circuit. The retained global graph has

$$c = 486330, \quad q = 45540, \quad R = c + q = 531870.$$

It merges 45706 of the previous 532036 addition nodes. Each retained node has exactly the same source support as one of the original common-point nodes. Every such span is positive definite: if all its triples contain  $i$ , then  $\sum_j u_j = 3u_i$  and  $\langle u, u \rangle = \sum_{j \neq i} u_j^2 > 0$  for  $u \neq 0$ . The preceding proposition therefore applies, including after cross-group sharing. The physical reversible roles are compiled afresh from this graph; they are not obtained by identifying simultaneously occupied scratch slots.

For completeness, if a graph has  $c$  additions and  $q$  output uses, it has  $2c + q$  outgoing uses in total. Input gates allocate one role per use, and each addition identifies its pivot input with one outgoing role. The resulting invertible gather-and-fanout product  $L$  uses  $c + q$  roles. With  $V$  the source copy,  $J$  the output injection, and  $G, R$  the retained central operations, use

$$L, J, L^{-1}, R, V, G, R, L, J, L^{-1}, G, V.$$

The two side injections sum to  $JLz + JL(z + Vx) = JLVx$ , and the final cleanup restores arbitrary scratch  $z$ . Centers are also restored. Since  $JLV + RG = I$ , each invocation is the original bank shear. The reversed invocation uses the reversed inverse schedule and the complement frames just proved.

The only decreasing transitions remain the original central returns, of total dimension  $L_{\text{dec}} = 3v^2h^2$ . Every auxiliary role starts at  $D_0$  and ends at  $D_1$ , so the previous bijective first/third-stage sharing, including centers, still applies without changing its joining labels. Thus

$$\begin{aligned} W_b^{\text{shared}} &= 2N + 2v^2(R + h) = 252137288620800, \\ s_b^{\text{shared}} &= W_b^{\text{shared}}m - N + 6v^2h^2 = 24542034552800107200, \\ W_b^{\text{shared}}m - s_b^{\text{shared}} &= 572394081600. \end{aligned}$$

**Proposition 2** (Cross-group shared-computation bit interface). *The shared-point bit network exchanges the scalar banks, restores arbitrary auxiliary inputs, and satisfies the original all-role endpoint identity with  $W_b^{\text{shared}}$  roles and total edge rank  $s_b^{\text{shared}}$ .*

*Proof.* The scalar and frame arguments above preserve the original stage boundaries. Forward, inverse, forward shears exchange the banks. All surviving auxiliary roles have endpoint matrices zero and  $I_m$  and are fixed by the exchange. Telescoping gives  $W_b^{\text{shared}}m - 2N + 2L_{\text{dec}}$  total absolute label change, and the unchanged rational source correction adds  $N$ . The original common-frame and rational-shear transfer applies; all graphs, bases, and gates are fixed finite data.  $\square$

## 1.2 Explicit rational bounds for the shared-point recurrences

Put  $a_b = 203/10^{11}$ ,  $a_c = 9/500000000000$ , and  $L_0 = 5743/500$ . The exact comparisons are

$$\eta_b^{\text{shared}} = \frac{27}{1157655136} > a_b L_0, \quad \eta_c = \frac{7}{22253827054} > a_c L_0.$$

Together with the retained logarithm enclosure  $\log m < L_0$  and  $e^{-x} > 1 - x$  for  $x > 0$ , these imply the required strict recurrence bounds with

$$\tau = 1 - \frac{203}{10^{11}}, \quad \sigma = 1 - \frac{9}{500000000000}. \quad (1)$$

## 2 Conditional exponent and checks

With  $a = 203/10^{11}$  use

$$\epsilon = 1/21, \quad \beta = 9/10, \quad c = 9a/10, \quad \lambda = 1 - 19a^2/20, \quad \lambda' = 1 - 9a^2/10, \quad \delta = 1/16, \quad C_1 = 20.$$

The existing exact constraint checker gives

$$\min_j g_j = \frac{3a^2}{70} = \frac{17661}{10^{23}} > \frac{13}{2^{66}}.$$

This strict margin absorbs the fixed logarithmic factors in the retained assembly argument. The resulting conditional bound is  $O(n(\log n)^{1-13 \cdot 2^{-66}})$ .

Every retained addition is checked by exact source-support comparison in local pair coordinates. Equal sums from different groups are decoded by their fixed pair and variable vertices. Both directions of frame inclusion are checked on every compiled physical role. Small complete invocations are independently expanded on every input basis vector, including every dirty-scratch input, and a full small three-stage model checks bank exchange and scratch sharing. The written complement-frame proof supplies the rational interpretation of these checks. They do not formally verify the upstream theorem.

The artifacts are `scripts/shared_point_circuit.py`, `scripts/shared_point_network.py`, `certificates/shared-point-network.json`, and `patches/h46-shared-point.patch`.

## 3 What this does and does not open up

For this particular finite graph, the retained necessary inequality  $\kappa < a^2/(20(1-a))$  and an exact upper enclosure of  $a = -\log(1-\eta)/\log m$  give a ceiling below  $2^{-62}$ . This is not a ceiling on cross-group sharing or on other finite graphs. With the same  $h = 46$  rank deficit and downstream recipe, a future graph using at most 158353 side roles would suffice for  $a = 6.4 \cdot 10^{-9}$  and  $\kappa = 2^{-59}$ . No such graph is constructed here.